

BUILD CANON

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Abstract

BUILD is governed DAG traversal phases are nodes, dependencies are edges, parallelism is discovered, GOV is the only hardcoded path.

hadleylab.org Governed Research. Every claim cited.

Contents

- **BUILD_IS_DAG** compilation is DAG traversal; phases are nodes and dependencies are edges discovered from the governed pipeline config.
- **PARALLELISM_IS_DISCOVERED** parallelism is derived from the DAG by topological sort, never hardcoded; phases glob-discover themselves.
- **GOV_BOUNDARY_GATED** the compiler gates the GOV boundary at compile time; the build FAILS if runtime files live in GOV.
- **GENERATED_MARKER_REQUIRED** every file written back to GOV carries a `_generated` marker so the boundary between authored and compiled is always visible. **## Constraints**

MUST: Gate GOV boundary at compile time build FAILS

MUST: Discover phases via glob (not hardcoded list)

MUST: Discover DAG edges from governed pipeline config

MUST: Use GovGraph (galaxy BFS) for all cross-scope

MUST: Carry `_generated` marker on files written back

MUST: Auto-scaffold missing closure artifacts from

MUST: Fail fast if pipeline config is missing or d

MUST: Three environments LOCAL, STAGING, PRODUCTION

MUST: LOCAL build passes all gates before STAGING

MUST: STAGING passes smoke tests before PRODUCTION

MUST: PRODUCTION deploy requires explicit promotion

MUST: build-verify runs as terminal gate no deploy

MUST: All paths discovered from GOV tree no hardcode

MUST NOT: Push to PRODUCTION without passing LOCAL + STAGING

MUST NOT: Hardcode scope paths, fleet paths, repo paths

MUST NOT: Write runtime artifacts into GOV tree

MUST NOT: Maintain fallback paths for missing galaxy

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